

MARIAZELL, Austria

WMO: 111720

Lat: 47.767N

Lon: 15.317E

Elev: 875

StdP: 91.25

Time Zone: 1.00 (EUC)

Period: 83-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-13.0	-20.7	0.7	-12.0	-17.4	0.9	-11.5	12.8	1.7	11.1	1.2	0.3	120	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	26.8	15.9	24.9	15.4	23.3	14.7	17.2	23.7	16.3	22.6	15.5	21.5	5.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	11.8	19.3	14.0	11.1	18.2	13.1	10.5	17.5	51.7	23.5	48.9	22.8	46.6	21.7	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	10.1	9.3	-20.1	29.9	4.6	1.6	-23.4	31.1	-26.1	32.0	-28.7	32.9	-32.0	34.0		
(5)			-20.5	19.3	4.5	1.5	-23.7	20.3	-26.4	21.2	-28.9	22.0	-32.1	23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-2.5	-1.1	1.5	5.6	11.2	13.3	15.6	15.8	11.5	8.0	2.0	-1.8		
	DBStd	7.72	4.77	4.73	4.37	3.96	3.74	3.54	3.13	3.30	3.51	4.06	4.45	4.55		
	HDD10.0	1861	387	309	265	142	31	8	1	2	23	87	242	364		
	HDD18.3	4292	645	543	523	382	223	154	93	91	204	321	490	623		
	CDD10.0	640	0	0	1	9	67	108	176	182	70	25	2	0		
	CDD18.3	28	0	0	0	0	1	5	10	12	1	0	0	0		
	CDH23.3	342	0	0	0	0	7	59	126	142	7	0	0	0		
	CDH26.7	52	0	0	0	0	0	7	19	25	0	0	0	0		
(14) Wind	WSAvg	2.3	1.9	2.4	2.5	2.7	2.5	2.3	2.3	2.1	2.1	2.7	2.3	2.0		
(15) Precipitation	PrecAvg	1106	65	63	87	71	110	129	141	116	105	73	75	72		
	PrecMax	1418	174	137	187	123	292	337	355	271	294	185	153	137		
	PrecMin	762	4	16	35	14	38	46	40	35	36	6	0	21		
	PrecStd	157	41	35	44	31	57	58	74	54	57	40	35	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	12.9	16.4	20.1	24.3	28.0	29.2	29.3	24.5	21.2	15.5	10.2		
		MCWB	3.8	5.4	7.8	10.7	13.8	16.0	17.2	16.4	14.8	12.2	9.0	5.0		
	2%	DB	7.3	9.4	13.2	17.4	22.4	25.2	26.7	26.9	22.3	18.6	12.7	7.9		
		MCWB	3.0	4.3	6.1	9.3	12.9	15.4	16.3	15.7	14.0	11.4	7.4	4.1		
	5%	DB	5.5	7.5	10.6	15.1	20.3	22.9	24.7	24.7	20.2	16.5	10.7	6.2		
		MCWB	2.1	3.1	4.9	8.1	12.0	14.5	15.5	15.4	13.3	10.5	6.1	3.3		
	10%	DB	3.8	5.4	8.0	12.8	18.4	20.7	22.7	22.8	18.3	14.6	8.8	4.3		
		MCWB	1.3	2.1	3.8	7.0	11.2	13.4	14.8	14.8	12.5	9.7	5.4	2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	6.2	8.4	11.9	15.4	17.9	19.2	18.1	16.2	13.6	9.8	6.2		
		MCDB	7.6	10.7	15.0	18.5	21.3	24.5	25.9	24.5	21.0	18.8	14.1	9.3		
	2%	WB	3.5	4.8	6.7	10.2	13.9	16.2	17.4	17.0	15.1	12.3	8.1	4.7		
		MCDB	6.5	8.5	11.9	15.8	20.5	23.1	23.9	23.6	20.7	17.0	12.0	6.9		
	5%	WB	2.5	3.5	5.4	8.7	12.7	15.1	16.4	16.1	14.1	11.3	6.8	3.5		
		MCDB	5.1	6.8	9.6	13.8	18.7	21.2	22.7	22.7	18.6	15.3	10.2	5.9		
	10%	WB	1.4	2.3	4.2	7.4	11.7	14.0	15.5	15.4	13.0	10.2	5.5	2.3		
		MCDB	3.5	5.0	7.5	11.9	17.2	19.7	21.2	21.4	17.3	13.9	8.3	4.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.0	7.9	8.5	9.6	9.2	9.6	10.0	8.9	8.6	6.8	6.1		
		MCDBR	9.0	11.4	13.5	13.3	13.4	13.9	14.4	14.5	13.3	12.8	9.9	8.2		
		MCWBR	5.8	6.8	7.4	6.7	6.3	5.8	6.1	5.6	6.3	6.6	5.9	5.6		
		MCWBR	8.0	9.9	11.9	11.3	12.1	12.3	12.0	12.3	11.4	10.3	8.5	7.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.283	0.311	0.351	0.391	0.395	0.400	0.405	0.401	0.373	0.342	0.313	0.283		
		taud	2.405	2.350	2.301	2.250	2.280	2.299	2.286	2.313	2.366	2.435	2.466	2.450		
		Ebn at Noon	802	849	863	859	868	864	853	840	831	803	758	759		
		Edh at Noon	67	88	109	127	128	126	126	118	102	81	64	58		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.14	2.02	3.05	4.17	4.82	5.27	5.21	4.66	3.36	2.26	1.24	0.94		
		RadStd	0.13	0.25	0.35	0.46	0.55	0.42	0.48	0.52	0.43	0.22	0.17	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (105 neighbors)	+0.80	N/A	N/A	+0.83	+0.45	+0.40	-104	-139	+84	+44

Nomenclature: See separate page